

Recap (September 11th)

- We defined Propositions
- We can build compound propositions out of simple propositions (In english this involves connectors)

Key words

1. AND
2. OR
3. NOT

Building blocks of an elegant theory

4. Implies
5. Bi-implies
6. XOR
7. NAND

Lecture

Example (on a compound propositions)

“If it is raining, then the grass is wet”

Both “**It is raining**” & “**The grass is wet,**” are propositions in their own right

We can use propositional variables

let “it is raining” = p

let “the grass is wet” = q

leaving the connectors in, our compound proposition is if p , then q

We represent this symbolically as $p \rightarrow q$

“ $\rightarrow$ ” Read as p implies q

If p and q are both propositions, then the conjunction of p with q is “ p and q are both true”

which we write as “ $p \wedge q$ ” and read as “ p AND q ”

The disjunction of p with q is “At least one of p or q is true”

written as “ $p \vee q$ ” and read as “ p OR q ”

this is the inclusive sense of OR, which we use as the default meaning.

We translate the exclusive or as “ $p \oplus q$ ” read as “ p XOR q ”

- All version of implication permit to conclude, q is true once we learn p is true, “If it is raining, then the grass is wet”

Sometimes there is a contextual extra conclusion possible,

Example “If you eat your veg, then you get dessert”

- There is an implied threat, no dessert means no veg, we can be explicit by saying “ p if and only if q ”

$p \leftrightarrow q$ read as “ p iff q ” or “ p if and only if q ”

Example

- Brute force computations can be carried out with Truth Tables

p	q	$\neg p$	$\neg q$	$p \wedge q$	$p \vee q$	$\neg p \wedge \neg q$	$\neg(p \wedge q)$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	F	T	T	F	F	F	T	T
T	F	F	T	F	T	F	T	T	F	F
F	T	T	F	F	T	F	T	T	T	F
F	F	T	T	F	F	T	T	F	T	T

if p is a proposition, we define the *negation* of p as the compound statement “ p is not true”

The sentence “if p, then q” translated as “ $p \rightarrow q$ ” and read as “p implies q”

[illegible]